

4.7.13. Let S be the statement: The product of any irrational number and any nonzero rational number is irrational.

- a. Write a negation for S .
- b. Prove S by contradiction.

- a. Write a negation for S .

$S =$ The product of any irrational number and any nonzero rational number is irrational.

$$\forall s \in \mathbb{R}, \forall r \in \mathbb{Q}, s \notin \mathbb{Q} \wedge r \neq 0 \rightarrow rs \notin \mathbb{Q}$$

$\sim S =$ There is an irrational number and a nonzero rational number, where the product is not irrational.

$$\exists s \in \mathbb{R} \text{ st } \exists r \in \mathbb{Q} \text{ st } s \notin \mathbb{Q} \wedge r \neq 0 \text{ st } rs \in \mathbb{Q}$$

Solution: $\exists s \in \mathbb{R} \text{ st } \exists r \in \mathbb{Q} \text{ st } s \notin \mathbb{Q} \wedge r \neq 0 \text{ st } rs \in \mathbb{Q}$

- b. Prove S by contradiction

Proof:

Suppose not. That is, suppose there is an irrational number and a nonzero rational number.

We must show that the product is irrational.

The definition of a rational number is:

$$r \in \mathbb{Q} \leftrightarrow \exists a, b \in \mathbb{Z} \text{ st } r = a/b \text{ and } b \neq 0$$

Let $s \notin \mathbb{Q}$

Let $r \in \mathbb{Q}, r \neq 0$

$$r = a/b$$

Let's assume $rs \in \mathbb{Q}$

$$s = rs/r$$

Since r is a rational number a/b , for some $a, b \in \mathbb{Z}$, and rs is a rational number, which can be written as $c, d \in \mathbb{Z}$, rs/r is also a rational number.

Because $s = rs/r$, which is rational, then $s \in \mathbb{Q}$.

But $s \notin \mathbb{Q}$, therefore the statement is proven by contradiction, which's what was to be shown.

Solution: The product of any irrational number and any nonzero rational number is irrational, by proof by contradiction.